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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/420,293	05/23/2019	LARS KUHNERT	84158295(65080-3424)	4840

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EXAMINER	
GREGORY, BERNARR E	

ART UNIT	PAPER NUMBER
3648	

NOTIFICATION DATE	DELIVERY MODE
06/04/2021	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

docket@b2iplaw.com

Office Action Summary	Application No. 16/420,293	Applicant(s) KUHNERT et al.	
	Examiner BERNARR E GREGORY	Art Unit 3648	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 23 May 2019.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 6-20 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 6-20 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☑ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 23 May 2019 is/are: a)☐ accepted or b)☑ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☑ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☑ All b)☐ Some** c)☐ None of the:

1.☑ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date 5/23/2019.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED NON-FINAL OFFICE ACTION

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Comments

2. In the information disclosure statement (IDS) of May 23, 2019, citations #2, #3, and #4 are lined through due to the lack of a date in each of these citations. Please see 37 CFR 1.98(b)(5).

3. In the claim listing in the preliminary amendment of May 23, 2019, the status identifier for cancelled claims 1-5 fails to comply with 37 CFR 1.121(c) in that it is not placed within parentheses.

4. It is noted that claims 1-5 stand CANCELLED.

5. The preliminary amendment of May 23, 2019 has been ENTERED.

6. The substitute specification has been ENTERED.

Objection to the Specification

7. The specification is hereby objected to under 37 CFR 1.71(a) due to a lack of clarity. First, the mention of “claim 1” in paragraph [0005] makes the text unclear in that

there is no pending claim 1. Claims should not be mentioned by specific number in the specification. Second, the mention of "subclaims" in paragraph [0005] is unclear in that there are no subclaims in U. S. practice.

Objections to the Drawings

8. The drawings are objected to under 37 CFR 1.83(a) because they fail to show the claim 1 "processor" and "memory"; the claim 16 "means for detecting a plurality of objects around a vehicle"; the claim 16 "means for developing a map ..."; the claim 17 "means for detecting objects in substantially equally spaced segments ..."; the claim 18 "means for smoothing the polygon by removing ..."; the claim 19 "means for extending one or more corner points of the polygon ..."; and, the claim 20 "means for developing the polygon based on at least one stationary or unconcealed object .... " Any structural detail that is essential for a proper understanding of the disclosed invention should be shown in the drawing. MPEP § 608.02(d). **Corrected drawing sheets in compliance with 37 CFR 1.121(d) are required in reply to the Office action to avoid abandonment of the application**. Any amended replacement drawing sheet should include all of the figures appearing on the immediate prior version of the sheet, even if only one figure is being amended. The figure or figure number of an amended drawing should not be labeled as "amended." If a drawing figure is to be canceled, the appropriate figure must be removed from the replacement sheet, and where necessary, the remaining figures must be renumbered and appropriate changes made to the brief description of the several views of the drawings for consistency. Additional replacement sheets may be necessary to show the renumbering of the remaining figures. Each drawing sheet submitted after the filing date of an application must be labeled in the top

margin as either "Replacement Sheet" or "New Sheet" pursuant to 37 CFR 1.121(d). If the changes are not accepted by the examiner, the applicant will be notified and informed of any required corrective action in the next Office action. **The objection to the drawings will not be held in abeyance**.

9. The drawings are objected to 37 CFR 1.84(o) because items I, II, III, IV, V, and VI in Figure 4 lack "suitable descriptive legends" that are necessary for understanding of the drawing, which suitable descriptive legends are **hereby required by the examiner for items I, II, III, IV, V, and VI in Figure 4**. **Corrected drawing sheets in compliance with 37 CFR 1.121(d) are required in reply to the Office action to avoid abandonment of the application**. Any amended replacement drawing sheet should include all of the figures appearing on the immediate prior version of the sheet, even if only one figure is being amended. The figure or figure number of an amended drawing should not be labeled as "amended." If a drawing figure is to be canceled, the appropriate figure must be removed from the replacement sheet, and where necessary, the remaining figures must be renumbered and appropriate changes made to the brief description of the several views of the drawings for consistency. Additional replacement sheets may be necessary to show the renumbering of the remaining figures. Each drawing sheet submitted after the filing date of an application must be labeled in the top margin as either "Replacement Sheet" or "New Sheet" pursuant to 37 CFR 1.121(d). If the changes are not accepted by the examiner, the applicant will be notified and informed of any required corrective action in the next Office action. **The objection to the drawings will not be held in abeyance**.

Claim Interpretation

10. The following is a quotation of 35 U.S.C. 112(f):

> (f) Element in Claim for a Combination. – An element in a claim for a combination may be expressed as a means or step for performing a specified function without the recital of structure, material, or acts in support thereof, and such claim shall be construed to cover the corresponding structure, material, or acts described in the specification and equivalents thereof.

The following is a quotation of pre-AIA 35 U.S.C. 112, sixth paragraph:

> An element in a claim for a combination may be expressed as a means or step for performing a specified function without the recital of structure, material, or acts in support thereof, and such claim shall be construed to cover the corresponding structure, material, or acts described in the specification and equivalents thereof.

11. The claims in this application are given their broadest reasonable interpretation using the plain meaning of the claim language in light of the specification as it would be understood by one of ordinary skill in the art. The broadest reasonable interpretation of a claim element (also commonly referred to as a claim limitation) is limited by the description in the specification when 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, is invoked.

As explained in MPEP § 2181, subsection I, claim limitations that meet the following three-prong test will be interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph:

(A) the claim limitation uses the term “means” or “step” or a term used as a substitute for “means” that is a generic placeholder (also called a nonce term or a non-structural term having no specific structural meaning) for performing the claimed function;

(B) the term "means" or "step" or the generic placeholder is modified by functional language, typically, but not always linked by the transition word "for" (e.g., "means for") or another linking word or phrase, such as "configured to" or "so that"; and

(C) the term "means" or "step" or the generic placeholder is not modified by sufficient structure, material, or acts for performing the claimed function.

Use of the word "means" (or "step") in a claim with functional language creates a rebuttable presumption that the claim limitation is to be treated in accordance with 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph. The presumption that the claim limitation is interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, is rebutted when the claim limitation recites sufficient structure, material, or acts to entirely perform the recited function.

Absence of the word "means" (or "step") in a claim creates a rebuttable presumption that the claim limitation is not to be treated in accordance with 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph. The presumption that the claim limitation is not interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, is rebutted when the claim limitation recites function without reciting sufficient structure, material or acts to entirely perform the recited function.

Claim limitations in this application that use the word "means" (or "step") are being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, except as otherwise indicated in an Office action. Conversely, claim limitations in this application that do not use the word "means" (or "step") are not being interpreted under

35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, except as otherwise indicated in an Office action.

In independent claim 16, "means for detecting a plurality of objects around a vehicle" and "means for developing a map" are interpreted as invoking 35 USC 112(f).

In dependent claim 17, "means for detecting objects in substantially equally spaced segments" is interpreted as invoking 35 USC 112(f).

In dependent claim 18, "means for smoothing the polygon by removing one or more corner points ..." is interpreted as invoking 35 USC 112(f).

In dependent claim 19, "means for extending one or more corner points of the polygon ..." is interpreted as invoking 35 USC 112(f).

In dependent claim 20, "means for developing the polygon based on at least one stationary or unconcealed object ..." is interpreted as invoking 35 USC 112(f).

12. The following is a quotation of 35 U.S.C. 112(b):

> (b) CONCLUSION.—The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the inventor or a joint inventor regards as the invention.

> The following is a quotation of 35 U.S.C. 112 (pre-AIA), second paragraph:
> The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the applicant regards as his invention.

13. Claims 6-20 are rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which the inventor or a joint inventor (or for applications subject to pre-AIA 35 U.S.C. 112, the applicant), regards as the invention.

Overall, independent claim 6 is indefinite and unclear in that the claim mentions "instructions" to perform the functions listed in the body of the claim, but the claim fails

to clearly and definitely recite the "instructions," and the "instructions" do not have clear antecedent in the disclosure. Please see 37 CFR 1.75(d)(1). Substantially the same remarks apply to each of dependent claims 7-10.

In each of claims 6-10, the uses of the term, "instructions" are indefinite and unclear in context in that there is no clear antecedent basis for the term in the specification. Please see 37 CFR 1.75(d)(1).

In claim 6, the term, "processor" is indefinite and unclear in context in that there is no clear antecedent basis for the term in the specification. Please see 37 CFR 1.75(1).

In claim 6, the term, "memory" is indefinite and unclear in context in that there is no clear antecedent basis for the term in the specification. Please see 37 CFR 1.75(1).

On line 3 of claim 6, the method step to "detect a plurality of objects around a vehicle" is unclear in context in that "a processor and a memory," even with "instructions," are not able to perform this method step, using the usual and ordinary meaning of the word "detect."

On lines 4-5 of claim 6, the method step, "develop a map of an environment around the vehicle based on a polygon having sides extending between the plurality of objects" is unclear in context, particularly as to what action is meant by "develop." What action is mean by the phrase to "develop a map"?

On lines 4-5 of claim 6, the phrase, "a polygon having sides extending between the plurality of objects" is indefinite and unclear in context. Is this phrase intended to mean that the "sides" connect the individual objects in the "plurality of objects"? Or, is this phrase intended to mean that the "sides" pass among the "plurality of objects"?

On line 2 of dependent claim 7, "the motor vehicle" lacks clear antecedent basis in that there is no earlier mention of a "motor vehicle" in either claim 7 or in claim 6.

On line 2 of independent claim 11, it is unclear what is meant in context by the method step, "detecting a plurality of objects around a vehicle" in that comparing lines 2-3 of dependent claim 16 to line 2 of claim 11, the action of "detecting" does not appear to be the action of the disclosed "sensor." Substantially the same problem occurs on line 3 of claim 6.

On lines 3-4 of claim 11, the method step, "developing a map of an environment around the vehicle based on a polygon having sides extending between the plurality of objects" is unclear in context, particularly as to what action is meant by "developing." What action is mean by the phrase to "developing a map"?

On lines 3-4 of claim 11, the phrase, "a polygon having sides extending between the plurality of objects" is indefinite and unclear in context. Is this phrase intended to mean that the "sides" connect the individual objects in the "plurality of objects"? Or, is this phrase intended to mean that the "sides" pass among the "plurality of objects"?

On line 2 of dependent claim 12, "the motor vehicle" lacks clear antecedent basis in that there is no earlier mention of a "motor vehicle" in either claim 12 or in claim 11.

On line 1 of dependent claim 15, it is unclear in context what action is meant by the phrase, "developing the polygon."

On line 3 of independent claim 16, it is unclear in context what action is meant by "detecting a plurality of objects." Since line 2 of claim 16 recites a "sensor," line 3 of claim 16 cannot be directed to the action of the "sensor."

On lines 5-6 of claim 16, the phrase, "developing a map of an environment around the vehicle based on a polygon having sides extending between the plurality of objects" is unclear in context, particularly as to what action is meant by "developing." What action is mean by the phrase to "developing a map"?

On lines 5-6 of claim 16, the phrase, "a polygon having sides extending between the plurality of objects" is indefinite and unclear in context. Is this phrase intended to mean that the "sides" connect the individual objects in the "plurality of objects"? Or, is this phrase intended to mean that the "sides" pass among the "plurality of objects"?

Dependent claim 17 is indefinite and unclear in context as to whether the "means for detecting objects in substantially equally spaced segments" is the same as the claim 16 "means for detecting a plurality of objects around a vehicle."

On line 2 of dependent claim 17, "the motor vehicle" lacks clear antecedent basis in that there is no earlier mention of a "motor vehicle" in either claim 17 or in claim 16.

On lines 1-2 of dependent claim 20, it is unclear in context what action is meant by the phrase, "developing the polygon."

Each of claim limitations "means for detecting a plurality of objects around a vehicle" and "means for developing a map" in claim 16; "means for detecting objects in substantially equally spaced segments" in claim 17; "means for smoothing the polygon by removing one or more corner points …" in claim 18; "means for extending one or more corner points of the polygon …" in claim 19; and, "means for developing the polygon based on at least one stationary or unconcealed object …" in claim 20 invokes 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph. However, the written description fails to disclose the corresponding structure, material, or acts for performing

the entire claimed function and to clearly link the structure, material, or acts to the function. The specification fails to disclose any corresponding structure, material, or acts for performing the entire claimed function for any of the listed means above. Therefore, the claim is indefinite and is rejected under 35 U.S.C. 112(b) or pre-AIA 35 U.S.C. 112, second paragraph.

Applicant may:

(a) Amend the claim so that the claim limitation will no longer be interpreted as a limitation under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph;

(b) Amend the written description of the specification such that it expressly recites what structure, material, or acts perform the entire claimed function, without introducing any new matter (35 U.S.C. 132(a)); or

(c) Amend the written description of the specification such that it clearly links the structure, material, or acts disclosed therein to the function recited in the claim, without introducing any new matter (35 U.S.C. 132(a)).

If applicant is of the opinion that the written description of the specification already implicitly or inherently discloses the corresponding structure, material, or acts and clearly links them to the function so that one of ordinary skill in the art would recognize what structure, material, or acts perform the claimed function, applicant should clarify the record by either:

(a) Amending the written description of the specification such that it expressly recites the corresponding structure, material, or acts for performing the claimed function and clearly links or associates the structure, material, or acts to the claimed function, without introducing any new matter (35 U.S.C. 132(a)); or

(b) Stating on the record what the corresponding structure, material, or acts, which are implicitly or inherently set forth in the written description of the specification, perform the claimed function. For more information, see 37 CFR 1.75(d) and MPEP §§ 608.01(o) and 2181.

14. The following is a quotation of the first paragraph of 35 U.S.C. 112(a):

> (a) IN GENERAL.—The specification shall contain a written description of the invention, and of the manner and process of making and using it, in such full, clear, concise, and exact terms as to enable any person skilled in the art to which it pertains, or with which it is most nearly connected, to make and use the same, and shall set forth the best mode contemplated by the inventor or joint inventor of carrying out the invention.

The following is a quotation of the first paragraph of pre-AIA 35 U.S.C. 112:

> The specification shall contain a written description of the invention, and of the manner and process of making and using it, in such full, clear, concise, and exact terms as to enable any person skilled in the art to which it pertains, or with which it is most nearly connected, to make and use the same, and shall set forth the best mode contemplated by the inventor of carrying out his invention.

15. Claims 6-20 are rejected under 35 U.S.C. 112(a) or 35 U.S.C. 112 (pre-AIA), first paragraph, as failing to comply with the written description requirement. The claim(s) contains subject matter which was not described in the specification in such a way as to reasonably convey to one skilled in the relevant art that the inventor or a joint inventor, or for applications subject to pre-AIA 35 U.S.C. 112, the inventor(s), at the time the application was filed, had possession of the claimed invention.

Each of claim limitations “means for detecting a plurality of objects around a vehicle” and “means for developing a map” in claim 16; “means for detecting objects in substantially equally spaced segments” in claim 17; “means for smoothing the polygon by removing one or more corner points …” in claim 18; “means for extending one or more corner points of the polygon …” in claim 19; and, “means for developing the

polygon based on at least one stationary or unconcealed object …" in claim 20 invokes 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph. However, the written description fails to disclose the corresponding structure, material, or acts for performing the entire claimed function and to clearly link the structure, material, or acts to the function. The specification fails to disclose any corresponding structure, material, or acts for performing the entire claimed function for any of the listed means above. Judging from a comparison of claims 16-20 with claims 6-10, it appears that the means of claims 16-20 would be performed by software or an algorithm; however, there is no disclosure of software or an algorithm in either the specification or in the drawings (e.g., Figure 4) to implement the specialized functions with each of the means in claims 16-20. Consequently, one of ordinary skill-in-the-art could not reasonably conclude that the inventors or applicant had possession of the invention as of the date of filing.

Due to the parallelism between the sets of claims, namely, claims 6-10, claims 11-15, and claims 16-20, one of ordinary skill-in-the-art could not reasonably conclude that the inventors or applicant had possession of the invention as set forth in claims 6-15 as of the date of filing.

Citation of Prior Art of General Interest

16. The prior art made of record and not relied upon is considered pertinent to applicant's disclosure.

Zou et al ('498) is of general interest for showing the use of a "bounding polygon."

Wodrich et al ('994) is of general interest for showing a vehicle sensor that uses a rectangular grid.

Brown et al ('421) is of general interest for showing a vehicle radar that uses a matrix, and for the hexagon in Figure 3.

Schaefer et al ('492) is of general interest for showing a system and method that maps obstacles in a rectangular grid, and for showing smoothing (e.g., see paragraph [0087]).

Hamada et al ('020) is of general interest for showing a vehicle radar that uses a rectangular grid.

Honda ('467) is of general interest for showing the use of vehicle sensors with a rectangular region.

Artebrant et al ('667) is of general interest for tracking of objects, and for the square grid in Figure 2.

Stiles et al ('471) is of general interest for showing sensors that search and area, and for Figures 5 and 6. In Figure 5, please note the unobstructed line-of-sight and the line-of-sight that is blocked by a hill as used in the square grid.

Wiemer et al ('437) is of general interest for showing a vehicle sensor system that stores information about objects surrounding the vehicle, and for the rectangular grid in Figure 8.

Zeng et al ('138) is of general interest for showing the use of a mesh grid with a vehicle sensor system.

Contact Information

Any inquiry concerning this communication or earlier communications from the examiner should be directed to BERNARR E GREGORY whose telephone number is (571)272-6972. The examiner can normally be reached on Mondays through Thursdays from 7:30 am to 5:30 pm eastern time.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Vladimir Magloire, can be reached at telephone number 571-270-5144. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see http://portal.uspto.gov/external/portal. Should you have questions about access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free).

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

/BERNARR E GREGORY/
Primary Examiner, Art Unit 3648